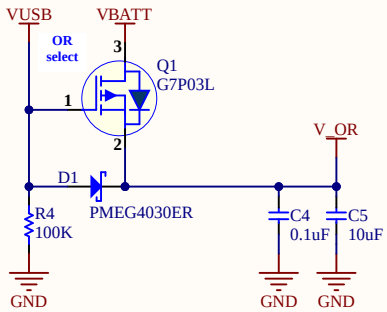
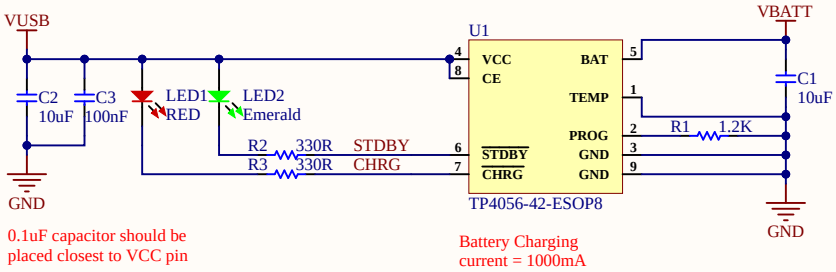
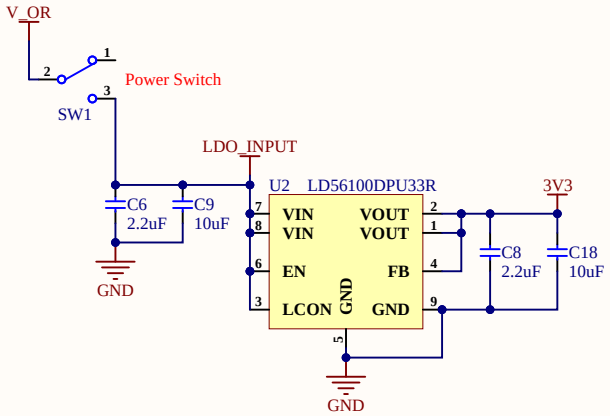
POWER SUPPLY ORING CIRCUIT



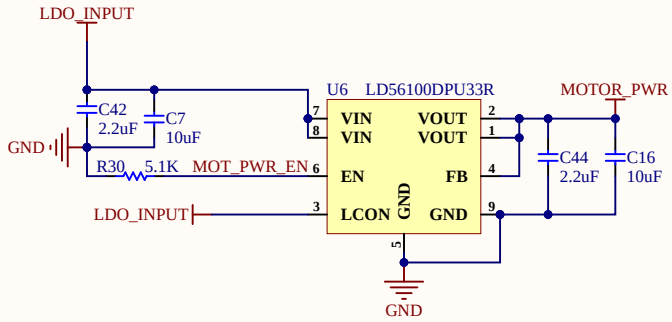
BATTERY CHARGING CIRCUIT



LDO_SECTION



MOTOR_PS_3.3V



MOT_PWR_EN — MOT_PWR_EN

V_OR — V_OR

VUSB — VUSB

VBATT — VBATT

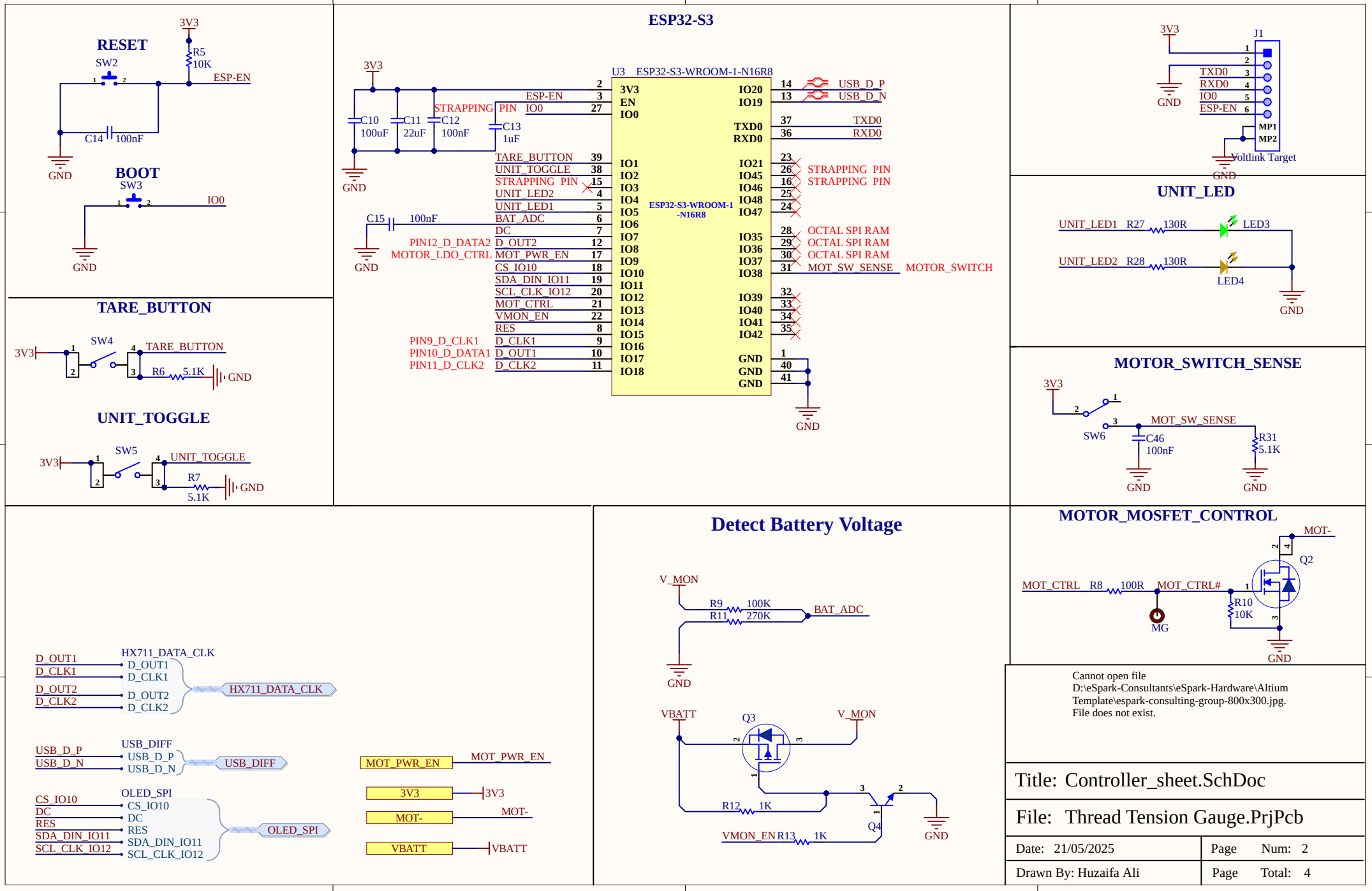
MOTOR_PWR — MOTOR_PWR

LDO_INPUT — LDO_INPUT

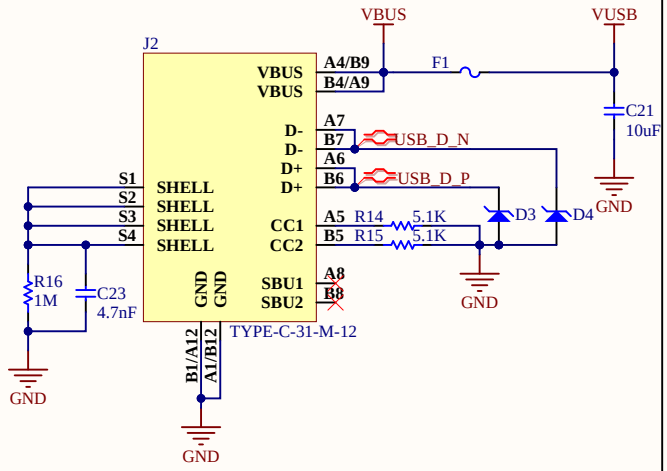
3V3 — 3V3

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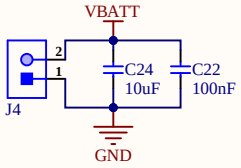
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File: Thread Tension Gauge.PrjPcb	
Date: 21/05/2025	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 4


RESET
SW2
3V3
R5
10K
ESP-EN
C14
100nF
GND
BOOT
SW3
IO0
GND
TARE_BUTTON
SW4
3V3
TARE_BUTTON
R6
5.1K
GND
UNIT_TOGGLE
SW5
3V3
UNIT_TOGGLE
R7
5.1K
GND
ESP32-S3
U3 ESP32-S3-WROOM-1-N16R8
3V3
EN
IO0
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8
IO9
IO10
IO11
IO12
IO13
IO14
IO15
IO16
IO17
IO18
ESP32-S3-WROOM-1-N16R8
IO20
IO19
TXD0
RXD0
IO21
IO45
IO46
IO48
IO47
IO35
IO36
IO37
IO38
IO39
IO40
IO41
IO42
GND
GND
GND
USB_D_P
USB_D_N
TXD0
RXD0
STRAPPING PIN
STRAPPING PIN
OCTAL SPI RAM
OCTAL SPI RAM
OCTAL SPI RAM
MOT_SW_SENSE
MOTOR_SWITCH
ESP-EN
STRAPPING PIN
IO0
C10
100uF
C11
22uF
C12
100nF
C13
1uF
C15
100nF
TARE_BUTTON
UNIT_TOGGLE
STRAPPING PIN
UNIT_LED2
UNIT_LED1
BAT_ADC
DC
PIN12_D_DATA2
D_OUT2
MOTOR_LDO_CTRL
MOT_PWR_EN
CS_IO10
SDA_DIN_IO11
SCL_CLK_IO12
MOT_CTRL
VMON_EN
RES
PIN9_D_CLK1
D_CLK1
PIN10_D_DATA1
D_OUT1
PIN11_D_CLK2
D_CLK2
3V3
J1
GND
TXD0
RXD0
IO0
ESP-EN
MP1
MP2
Voltlink Target
GND
UNIT_LED
UNIT_LED1
R27
130R
LED3
UNIT_LED2
R28
130R
LED4
GND
MOTOR_SWITCH_SENSE
3V3
SW6
MOT_SW_SENSE
C46
100nF
R31
5.1K
GND
GND
MOTOR_MOSFET_CONTROL
MOT-
Q2
MOT_CTRL
R8
100R
MOT_CTRL#
R10
10K
MG
GND
Detect Battery Voltage
V_MON
R9
100K
R11
270K
BAT_ADC
GND
VBATT
Q3
V_MON
R12
1K
VMON_EN
R13
1K
Q4
GND
D_OUT1
D_CLK1
D_OUT2
D_CLK2
HX711_DATA_CLK
D_OUT1
D_CLK1
D_OUT2
D_CLK2
HX711_DATA_CLK
USB_D_P
USB_D_N
USB_DIFF
USB_D_P
USB_D_N
USB_DIFF
CS_IO10
DC
RES
SDA_DIN_IO11
SCL_CLK_IO12
OLED_SPI
CS_IO10
DC
RES
SDA_DIN_IO11
SCL_CLK_IO12
OLED_SPI
MOT_PWR_EN
MOT_PWR_EN
3V3
3V3
MOT-
MOT-
VBATT
VBATT
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Date: 21/05/2025
Page Num: 2
Drawn By: Huzaifa Ali
Page Total: 4

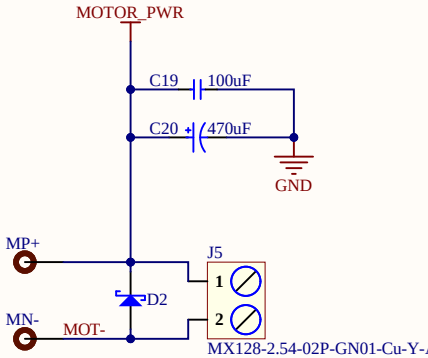
USB-C_INPUT



BATTERY INPUT

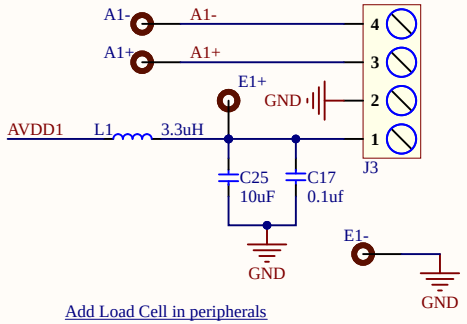


MOTOR CONNECTOR



Add 3V (101RPM) Gear Brushed DC Motor in peripherals

LOAD CELL CONNECTOR

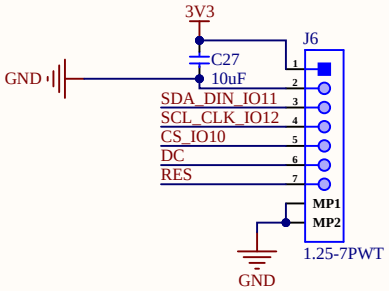


Add Load Cell in peripherals

LEDs



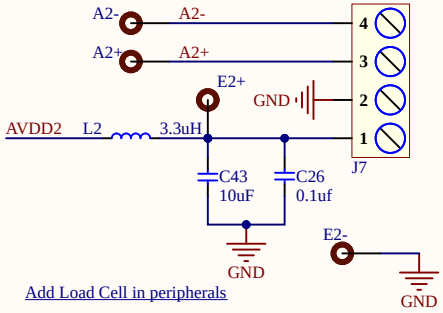
OLED CONNECTOR



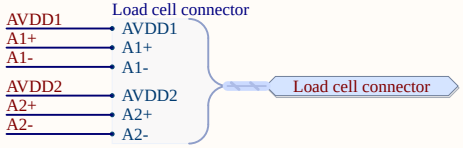
Add OLED Display in peripherals

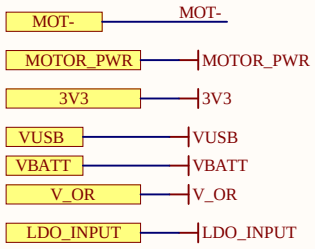
Add 7 Pin 15cm Double head inter-connection cable in peripherals

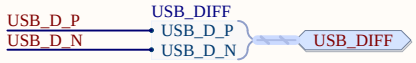
LOAD CELL CONNECTOR



Add Load Cell in peripherals





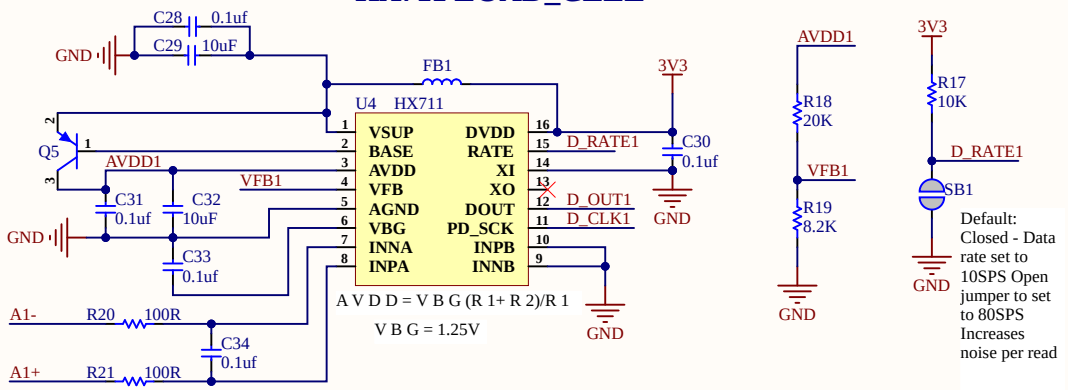




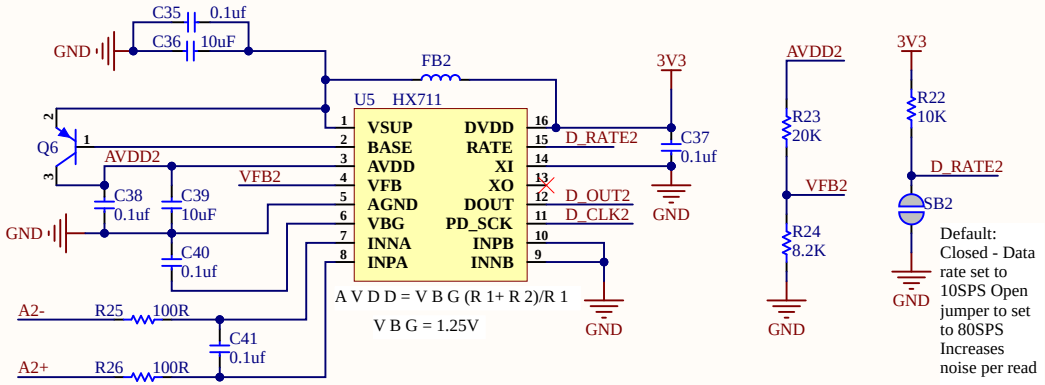
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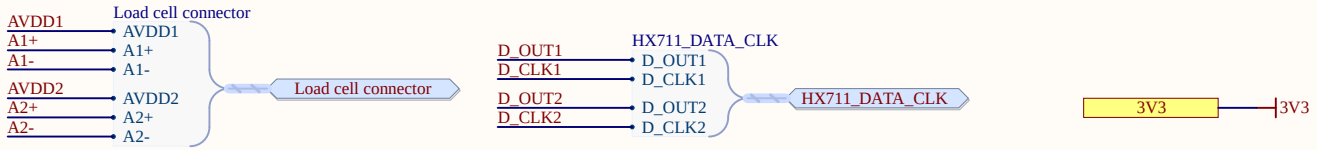
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File: Thread Tension Gauge.PrjPcb	
Date: 21/05/2025	Page Num: 3
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HX711 LOAD_CELL



HX711 LOAD_CELL





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Title: Load cell_Sheet.SchDoc	
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Date: 21/05/2025	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 4